Connect the car

Read the named terminals here; page 2 locates the actual solder pads. All grounds connect together. Crossings connect only where marked with a dot.

1 BATTERY

2S / 7.4 V / 300 mAh

8.4 V when fully charged

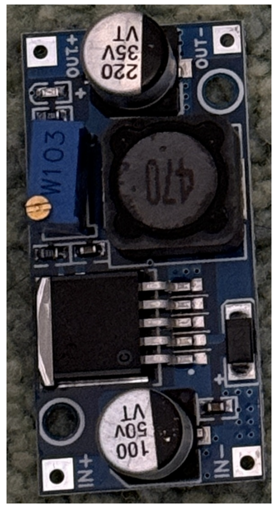
XT30
unplug = OFF

+
-

FUSE
4 A*

IN+
IN-

2 YOUR LARGE BUCK



Adjust output to 5.00 V

OUT+ / 5 V BUS

OUT- / GND BUS

3 FRONT-LEFT SERVO

RED +
BROWN -
S

Owned continuous-rotation unit

GPIO16 > LEFT SIGNAL

GPIO18 > RIGHT

4 REAR-RIGHT SERVO

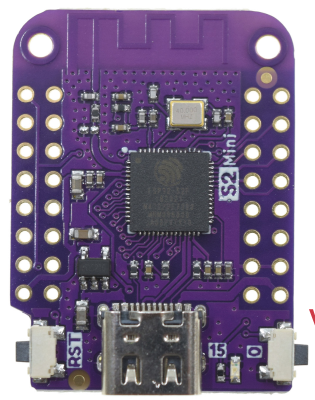
RED +
BROWN -
S

Same supply; separate signal

At the 5 V / GND split

Add **220 uF / 10 V electrolytic** (+ to 5 V; striped negative to GND), plus **100 nF ceramic** across the same rails. Keep leads short. Preserve the buck's onboard capacitor.

5 LOLIN S2 MINI



VBUS
J_PWR
JP
GND

USB unplugged while driving

Battery sensing: included on page 4

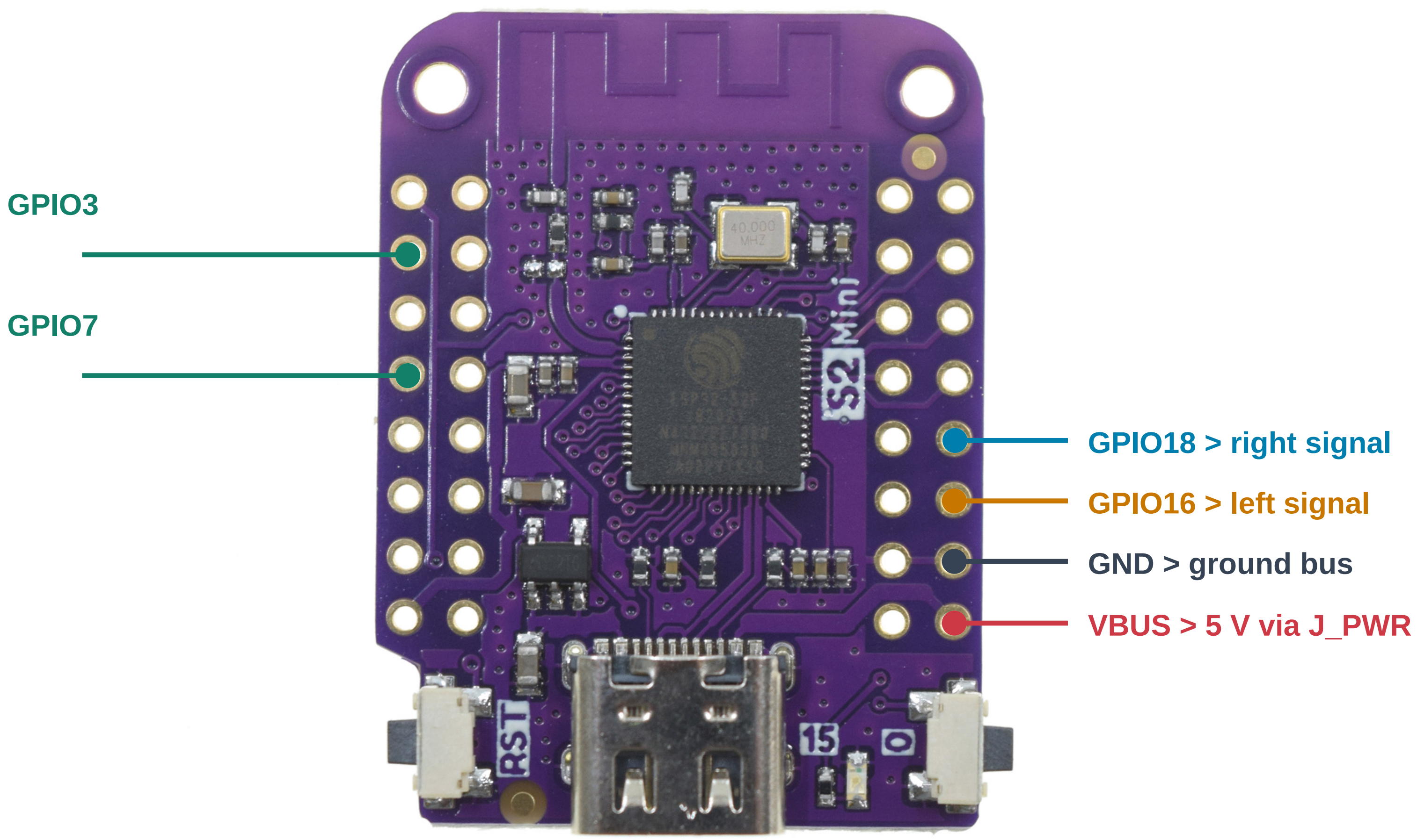
The existing receiver firmware also needs **GPIO3 + GPIO7** and the switched divider. It stops motion at 7.0 V; it does **not** disconnect the pack. Unplug after stopping.

* Fuse rating is provisional. Photos show reference servos; wire functions, not plug orientation, define the connections.

Exactly where each wire lands

Component side facing you. S2 antenna at the top; USB-C at the bottom. Photos are not mirrored.

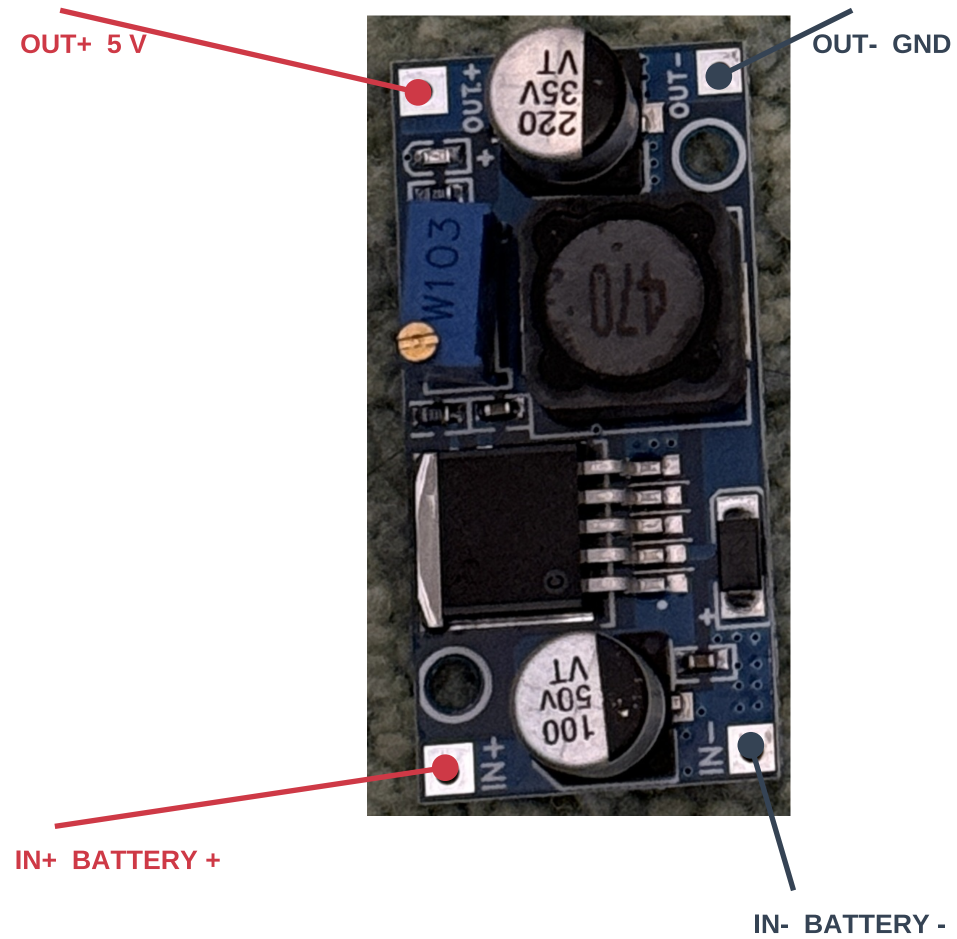
LOLIN S2 MINI v1.0.0 REFERENCE



Left GPIO3 / GPIO7: sensing only (page 4)

Use GPIO labels. Inner-row GPIO17 is NOT the right servo pin.

YOUR LARGE BUCK: SAME ORIENTATION



Blue trimmer sets output. Measure OUT+ to OUT- and set **5.00 V with all loads disconnected**. IC marking is not readable: confirm the board type.

Soldered wiring + removable connectors

Bare S2 holes and converter pads need soldered wires or soldered headers. Use 22 AWG or thicker for the battery and main 5 V / GND paths. Servo red normally means supply; brown/black ground; orange/yellow signal. Verify your servo wires before fitting a 3-pin extension. Split its three wires to the correct nets - do not plug the whole servo lead onto random adjacent S2 pins.

Battery, plug and converter choice

Reuse the larger buck for this prototype. A custom power PCB is not required.

BUY: LUMENIER 300

2S 75C LiPo, XT30
48 x 17 x 12 mm
18 g; 7.4 V nominal
8.4 V fully charged

$13.49; listed available
at the research snapshot.

RaceDayQuads product / purchase page

YOUR YELLOW CONNECTORS

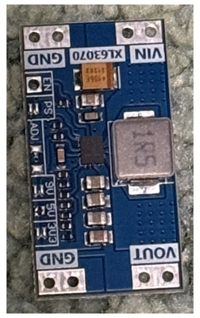
The photo establishes the XT family, but does not distinguish **XT30 from XT60**. Read the molded marking.

XT30: use the mating harness half.
XT60: buy an XT30 mating pigtail; a bulky adapter also works but increases footprint.

Check + / - with a meter. Housing shape or wire color alone is not proof of polarity.

Why the large buck wins here

A 2S pack feeds a step-down converter naturally. Allowing 1.2 A per servo plus 0.4 A for the S2 gives a **provisional 2.8 A at 5 V**. Actual servo peaks are unknown. The LM2596 IC is rated 3 A with suitable parts and cooling; this particular board must still pass load and temperature checks at 8.4 V and 7.0 V.



Why not the small XL63070 board?

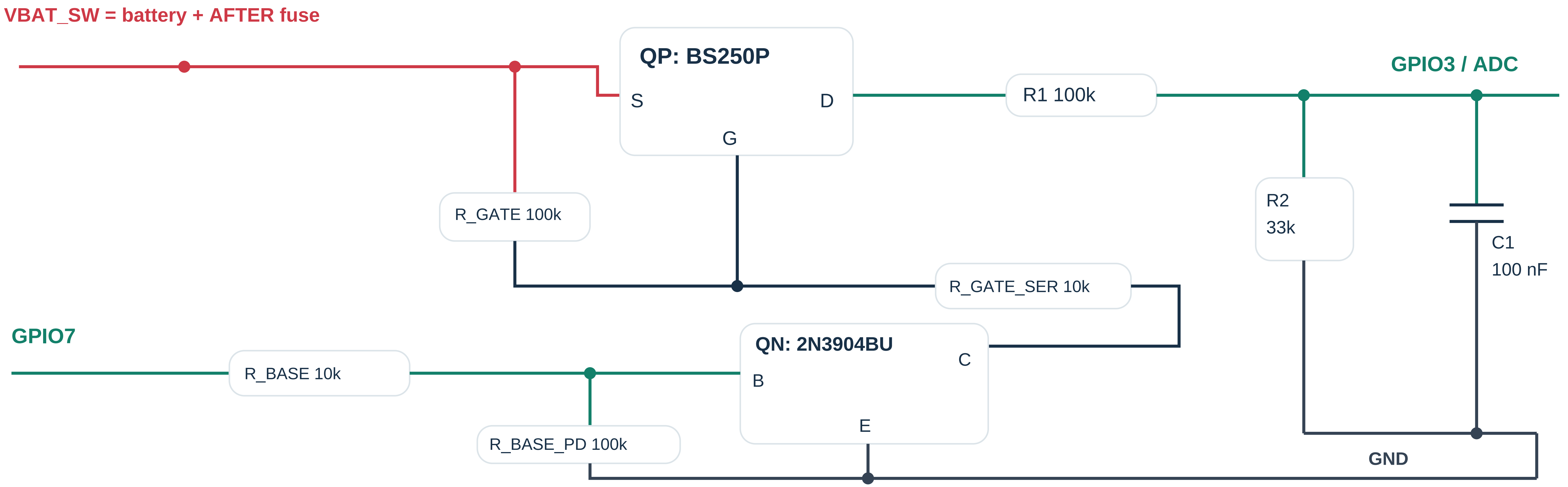
It appears to be a TPS63070-style buck-boost module; the IC is unconfirmed. TI specifies 2 A output at 4 V in / 5 V out. That is below our provisional load. A 1S design at 14 W could also need about 5 A near 3.3 V. Smaller hardware needs current measurements first.

Battery handling and fit

12 mm is the pack thickness, not a promised installed height: leave room for padding and lead bends. This hobby pack has no verified protection PCB. Neither owned converter provides charging, balancing or a suitable LiPo cutoff. Use a 2S balance charger and unplug the pack after each run.

Battery sensing for the existing firmware

A small insulated perfboard carries these low-current parts; motor current stays in the main harness.



Parts and terminal identification

QP: Diodes BS250P P-channel MOSFET: source to raw fused battery, drain to R1, gate to the resistor network shown.
QN: onsemi 2N3904BU NPN: emitter to GND, collector to R_GATE_SER, base to R_BASE.
Resistors: 3 x 100k, 2 x 10k, and 1 x 33k; use 1% for R1/R2. C1: 100 nF ceramic, 16 V or higher.
Match the exact maker's package-view drawing before soldering; transistor terminal letters are not left-to-right lead positions.

What it does - and what it cannot do

GPIO7 HIGH enables sensing; wait at least 20 ms before sampling GPIO3. Return GPIO7 LOW afterward.
8.4 V pack = about 2.084 V at GPIO3.
The switching stage avoids a sustained battery-fed ADC path when the S2 is off; do not replace it with a permanently connected divider.
Existing firmware needs pairing, servo-neutral and battery calibration before arming. Its 7.0 V stop only commands neutral. Pack voltage does not reveal an individually weak cell; monitor both cells and unplug promptly.

Build, check and use

No custom PCB. This is a soldered prototype harness, not a tested plug-and-play electrical assembly.

Wire and check in this order

1. Battery unplugged: solder the mating XT30 lead through a provisional 4 A inline fuse to IN+; negative to IN-. Insulate every joint.

2. Leave all loads disconnected. Power the buck, read OUT+ to OUT- with a multimeter and adjust to 5.00 V. Unplug before adding wires.

3. Split OUT+ and OUT- into separate servo and S2 branches. Add the capacitors. Fit J_PWR as a removable insulated connector in the S2 VBUS feed.

4. Add the page 4 sensing circuit. Check the ADC voltage and off-state behavior before connecting GPIO3. Calibrate against a meter.

5. With wheels lifted, test each servo, then both together. Check 5 V stability, resets and board temperature at 8.4 V and 7.0 V input. Do not hold a servo stalled.

6. Verify neutral on signal loss and the calibrated 7.0 V stop. Test voltage thresholds with an adjustable supply, not by deeply discharging the battery.

Before connecting USB to the S2

Unplug the battery, remove J_PWR, and disconnect both servo signal wires. Ground may remain connected. The S2 VBUS header is connected to USB power: battery disconnection alone can still let USB feed the servos or buck backward through the 5 V wire. Restore J_PWR and signals only after USB is removed.

Charge outside the car

Use a **2S LiPo balance charger** set for 4.20 V/cell (8.40 V pack). A conservative 1C setting for 300 mAh is **0.30 A**, subject to pack instructions. Connect its XT30 main lead and 3-pin balance lead as the charger requires. A 1S USB/TP4056 charger is not compatible. Charge attended on a nonflammable surface.

Run-time rule: use a 2S-compatible per-cell checker/alarm; around 3.5 V/cell under load is a conservative starting alarm. Stop and unplug promptly when it alarms or firmware stops. An alarm is not a cutoff. Remove both the XT30 and any balance-port alarm for storage. A real automatic cutoff requires additional suitable hardware.

Sources, image credits and build files

WEMOS: S2 Mini photo, board pinout and schematic

TI: LM2596 buck datasheet

TI: TPS63070 buck-boost datasheet

TowerPro: MG90S reference image (not proof of continuous rotation)

Diodes: BS250P terminal drawing

onsemi: 2N3904BU terminal drawing

Battery photo: Lumenier / RaceDayQuads (page 3). Converter and connector photos: user. Third-party images are not MIT-licensed.